```python

import numpy as np
from math import gamma, sin, pi

def fitness(x):
    return np.sum(x ** 2)

def levy_flight(dim, beta=1.5):
    sigma = (
        gamma(1 + beta) * sin(pi * beta / 2)
        / (gamma((1 + beta) / 2) * beta
           * 2 ** ((beta - 1) / 2))
    ) ** (1 / beta)

    u = np.random.normal(0, sigma, dim)
    v = np.random.normal(0, 1, dim)

    return u / (np.abs(v) ** (1 / beta) + 1e-10)

def cuckoo_search(n=25, dim=5, pa=0.25,
                  iterations=100, lower=-5, upper=5):

    nests = np.random.uniform(lower, upper, (n, dim))
    scores = np.array([fitness(x) for x in nests])

    best_idx = np.argmin(scores)
    best = nests[best_idx].copy()
    best_score = scores[best_idx]

    for _ in range(iterations):

        i = np.random.randint(n)
        new = nests[i] + 0.01 * levy_flight(dim)
        new = np.clip(new, lower, upper)
        new_score = fitness(new)

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            j = np.random.randint(n)
            if new_score < scores[j]:
                nests[j] = new
                scores[j] = new_score

            count = max(1, int(pa * n))
            worst = np.argsort(scores)[-count:]

            nests[worst] = np.random.uniform(
                lower, upper, (count, dim)
            )
            scores[worst] = np.array(
                [fitness(x) for x in nests[worst]]
            )

            idx = np.argmin(scores)
            if scores[idx] < best_score:
                best = nests[idx].copy()
                best_score = scores[idx]

        return best, best_score

    np.random.seed(42)
    solution, value = cuckoo_search()

    print("Best Solution:", solution)
    print("Best Fitness:", value)
```

```
PS C:\Users\BMSCECSE-L3-54> python -u "C:\Users\BMSCEC~1\AppData\Local\Temp\te
Best Solution: [-0.3054495  -0.51881766 -0.23813407  0.89185705  0.34170377]
Best Fitness: 1.3313494651494813
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```